⚙️ Configuration

Compute Device: `CPU Execution`

Embedding Model: `all-MiniLM-L6-v2` (384-d)

Classifier: `Calibrated Hybrid Tree Ensemble`

🎯 Risk Thresholds

Hallucination Risk Threshold

0.50

High Risk Flag Threshold

0.75

🧪 Demo Scenarios

Select a preconfigured test scenario:

2. Clearly Hallucinated (Expected High Risk)

> 📊 Benchmark Highlights

🔍 Claim-Level Hallucination Detector

Efficient hallucination detection in LLM answers using compact semantic embeddings and linguistic features.

User Question:

Who discovered penicillin?

Optional Reference / Context (leave blank for zero-context detection):

Sir Alexander Fleming was a Scottish physician and microbiologist who discovered penicillin in 1928.

LLM Generated Answer:

Marie Curie discovered penicillin in 1928 while conducting radioactivity research in Paris.

🚀 Detect Hallucinations

📊 Overall Answer Assessment

Overall Hallucination Risk	Peak Claim Risk	Total Claims Identified	High-Risk Claims
91.4%	91.4%	1	1 (100.0%)
↑ Suspicious	↑ High Risk		

📑 Claim-by-Claim Verification

Each sentence is independently extracted, parsed for linguistic and semantic features, and evaluated.

Claim 1 ⚠ Likely Hallucination (Risk: 91.4%)

"Marie Curie discovered penicillin in 1928 while conducting radioactivity research in Paris."

Important Model Signals:

• High named-entity specificity (3 entities) with elevated semantic tension

• Specific chronological assertion (year detected)

• Corroborated by provided reference context (similarity: 0.60)

> Inspect Features (Claim 1)

📌 **Important Disclaimer:** Model signals are indicators used by the detector and are not independent proof that a claim is factually false.